

Your name:

Group member names:

Analyze the data in R to complete the tables below. Answer the questions as you go.

Table 1. Analysis of existing experimental design.

Potential confounding variable	Difference in means	95% confidence interval (lcl, ucl)	Probability of the difference in means being due to sampling error (= random chance)

Q1. If individuals were randomly assigned to receive the therapy or the placebo, what would you expect the difference in means to be for any potential confounding variable?

Expected difference in means of confounding variables for randomized study = **0**

Q2. Is the value you wrote in the line above a plausible outcome for any of the potentially confounding variables in this study? Yes / **No** (circle one). Which piece of evidence allows you to answer this question? **Confidence intervals - none include 0 within their range.**

Q3. Overall, how would you describe the probability that the observed differences in means were just due to chance? **Very low (small p-values)**

Q4. Based on all of the information in Table 1, state whether or not individuals were randomly assigned to treatment groups. Individuals were not randomly assigned: **statistically significant differences in means between groups for all variables: CIs do not include 0, p vals are tiny**

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Complete Table 2 after randomizing treatment assignments in R.

Table 2. Analysis of modified experimental design.

Potential confounding variable	Difference in means	95% confidence interval (lcl, ucl)	Probability of the difference in means being due to sampling error (= random chance)

Q5. Use the information in Table 2 to assess whether assignment of individuals to treatment groups has been adequately randomized. Write out your assessment in the space below, referring to all three pieces of evidence (difference in means, 95% CIs, and p-values) in your explanation.

Randomization worked: diffs in means close to 0, CIs include 0 within their ranges, p-values are large (probability of observed diffs in means being due to chance is very high).